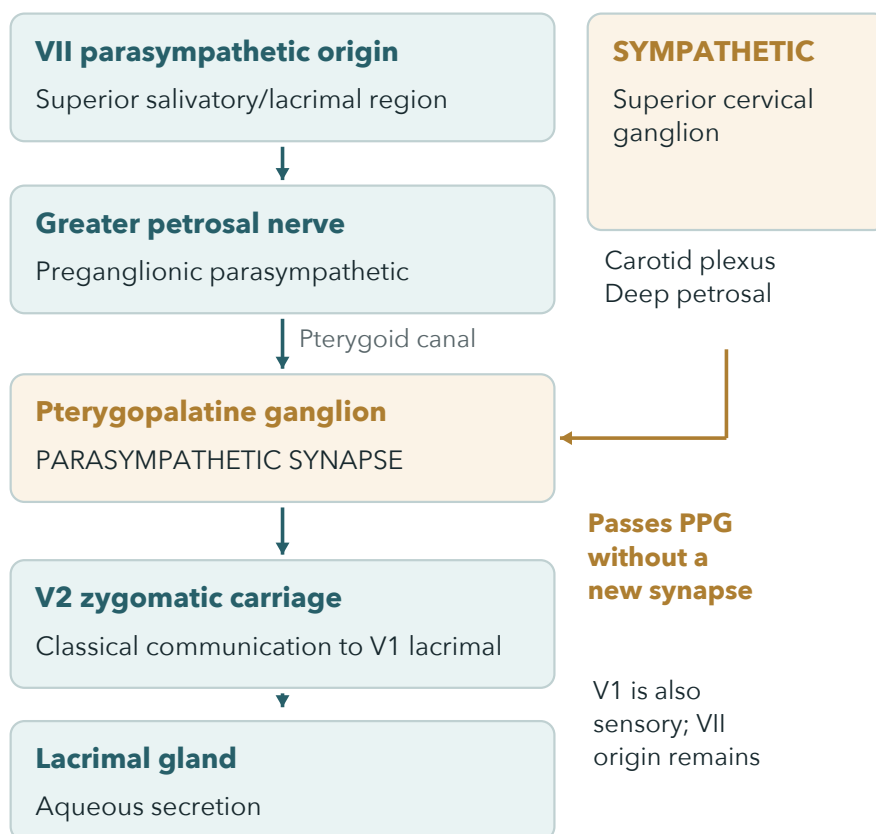


Lacrimal secretion and drainage

Production is a neural pathway; outflow is a passage from the medial lid to the nose.

The secretomotor route

Facial origin, trigeminal carriage



Classical lacrimal route. Postganglionic paths can also enter the gland directly; V2/V1 carriage does not change the parasympathetic origin.

Do not mix these fibres

Function	Key route
Parasympathetic secretion	VII; synapse in pterygopalatine ganglion.
Sympathetic vasomotor	Superior cervical ganglion; passes PPG without synapsing.
Gland / surface sensation	Trigeminal sensory pathways, including V1.

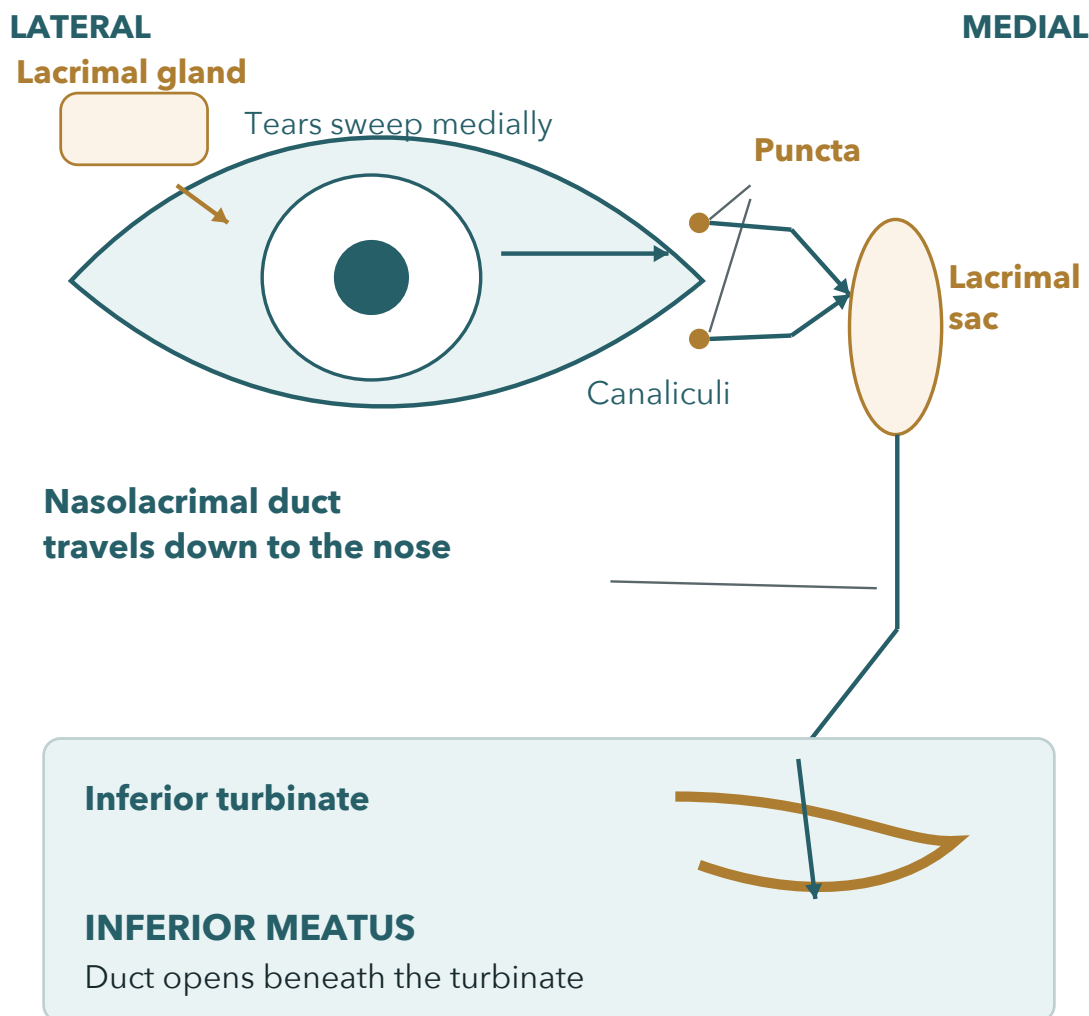
Read the selective deficit

Reduced tear output with intact corneal sensation and lid closure can reflect secretomotor injury. The blink afferent and facial motor output are separate components.

Sources NIH: Lacrimal gland | NIH: Petrosal pathways

The outflow route

From the surface to the inferior meatus



Orientation is labelled lateral to medial. Blinking distributes tears and helps the lacrimal drainage pump.

A blocked route, normal production

Observation	Localisation clue
Dye enters puncta and sac	The upstream canaliculi are patent.
No dye reaches the nose	A downstream nasolacrimal obstruction is possible.
Distal duct endpoint	Inferior meatus, beneath inferior turbinate.

The endpoint trap

The nasolacrimal duct drains into the inferior meatus. The middle meatus is a major paranasal sinus drainage region, not the normal lacrimal endpoint.

Sources NIH: Nasolacrimal anatomy | NIH: Lacrimal gland